

Smart Application in Artificial Intelligence and Machine Learning: Study and Understanding

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Student: To be completed

Executable desktop app, report, presentation and English video

6

ML modules

.exe

desktop delivery

Presentation roadmap

1. Foundation

AI and machine learning basics.

2. Workflow

how each module runs an experiment.

3. Architecture

interface, algorithms, visualization and packaging.

4. Demonstration

screenshots and module results.

5. Methods and applications

algorithms, domains and limits.

6. Defense plan and conclusion.

Close the project story and prepare for questions.

The project must prove execution, understanding and communication.

Executable Windows application required for validation.
Report title and structure follow the official PDF.
Presentation stays concise for a maximum 7-minute defense.
English video script is prepared for the final package.

Validation

Executable
Report
Presentation

AI and machine learning convert data into useful decisions.

AI combines data, algorithms, models, evaluation and deployment.

Machine learning learns patterns instead of hard-coding every rule.

The project uses random data to make experiments controlled and repeatable.

Core idea

Data

Model

Decision

Each module follows one clear scientific workflow.

Choose parameters with sliders.

Generate random data using
`numpy.random.uniform`.

Train the model and calculate metrics.

Interpret the graph and the result together.

Machine Learning workflow implemented in the application



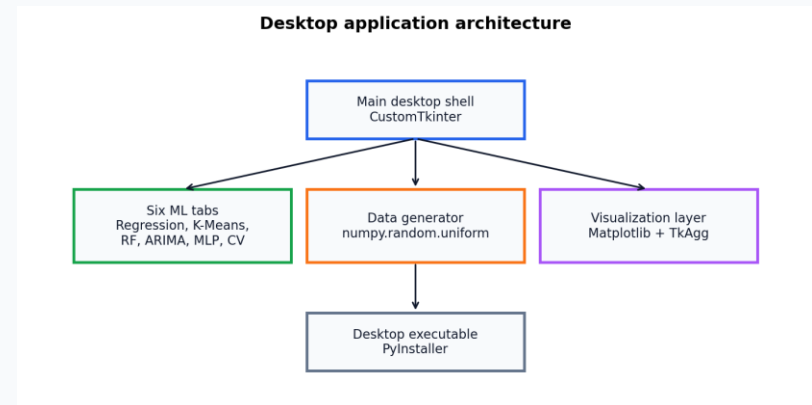
The application is modular and packaged as a desktop executable.

CustomTkinter provides the interface.

scikit-learn and statsmodels provide the algorithms.

Matplotlib displays embedded visualizations.

PyInstaller generates the final .exe.

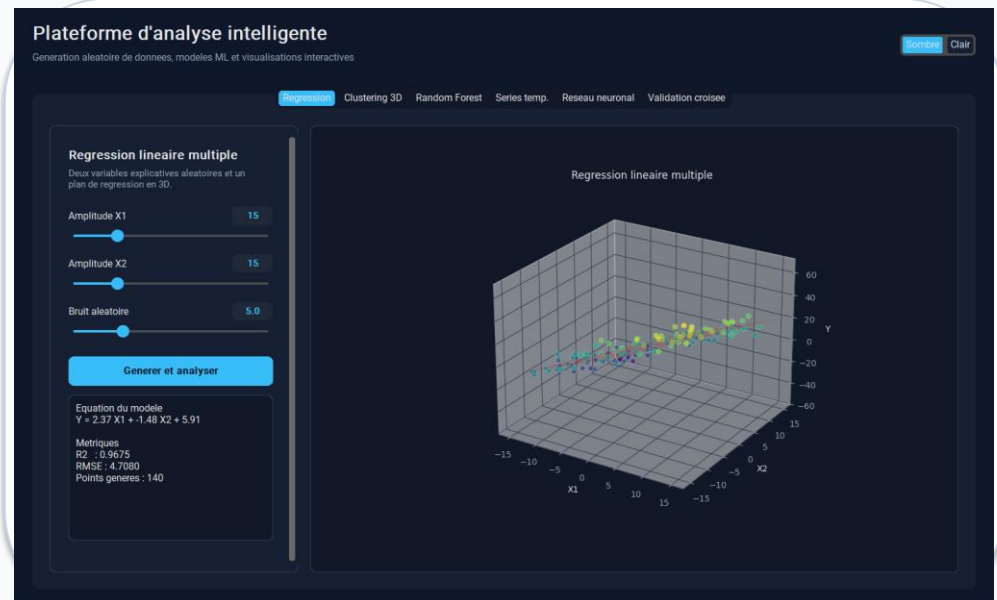


The regression tab demonstrates a complete ML experiment.

The user controls input ranges and noise.

The model estimates coefficients and intercept.

R2 and RMSE explain model quality.

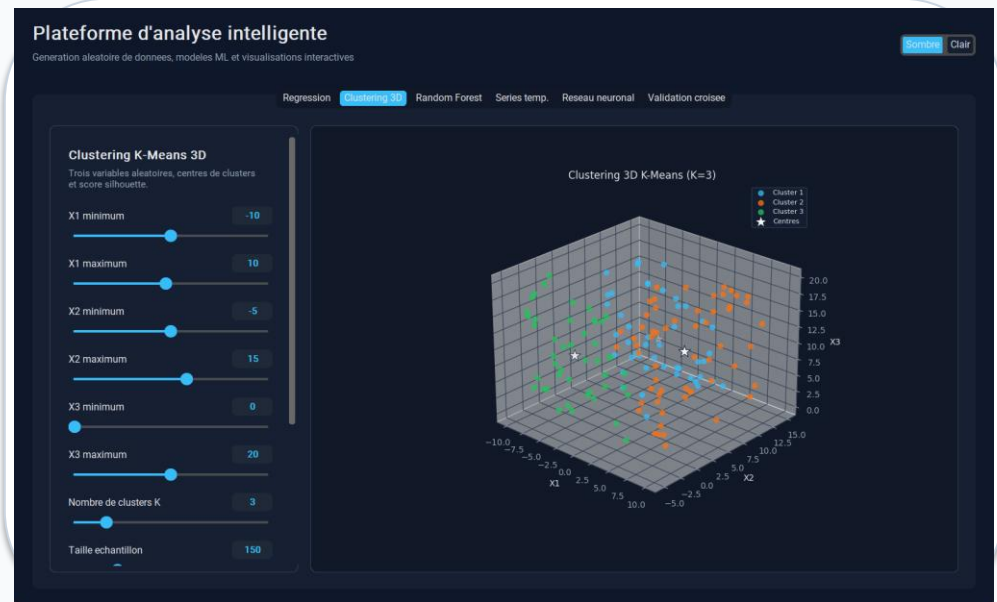


The clustering tab shows unsupervised learning in 3D.

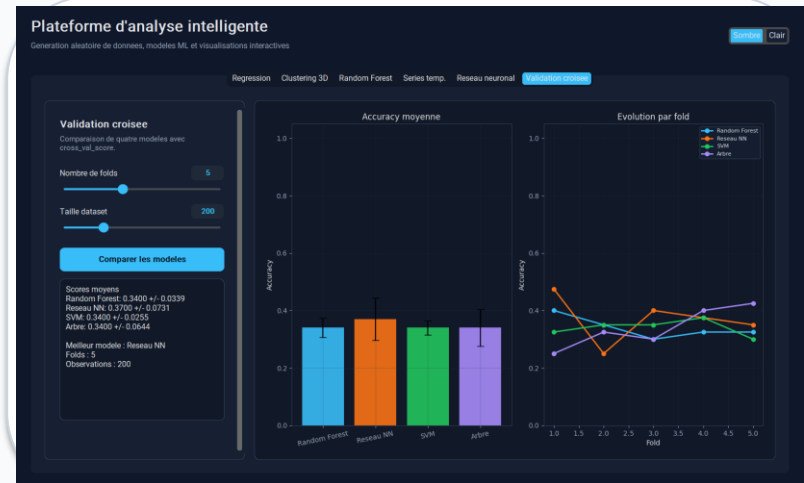
K-Means discovers groups without labels.

The 3D view shows separation between clusters.

The silhouette score summarizes cluster quality.



Classification and validation show performance comparison.



Random Forest explains accuracy and feature importance.

Cross-validation reduces dependence on one split.

The application compares several models in one place.

The six tabs cover the required machine learning families.

Regression

continuous prediction.

Clustering 3D

unsupervised grouping.

Random Forest

ensemble classification.

ARIMA, neural network and cross-validation

ARIMA, neural network and cross-validation complete the TP.

Defense focus

Explain what it does

Read the metric

Connect to the report

The report also studies additional algorithms.

SVM, Naive Bayes, Gradient Boosting and PCA.

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DBSCAN, Apriori, Hidden Markov Models and Autoencoders.

DBSCAN, Apriori, Hidden Markov Models and Autoencoders.

Each method is explained with use cases, strengths and limits.

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Defense focus

Explain what it does

Read the metric

Connect to the report

The same ML logic applies to three options.

Development: bug prediction, logs, DevOps monitoring.

Cybersecurity: anomaly detection, phishing and threat classification.

Artificial Intelligence: prediction, segmentation and decision support.

Three options

Development

Cybersecurity

AI

Machine learning needs evaluation and human interpretation.

Random data is useful for learning but less realistic than production data.

Overfitting, drift and poor data quality can reduce reliability.

Metrics must be interpreted with graphs and domain knowledge.

Risks

Data

Overfitting

Drift

The oral presentation is timed to stay under seven minutes.

- 1 minute: context, objective and AI/ML foundation.
- 3 minutes: application workflow, architecture and screenshots.
- 2 minutes: methods, domains, limits and perspectives.
- 1 minute: conclusion and questions.

Target

15 slides

About 28 sec each

Max 7 minutes

The final project is a complete academic and professional package.

The report proves scientific understanding.

The executable proves implementation.

The presentation and video prove communication.

GitHub or LinkedIn publication turns it into a portfolio asset.

Final message

Research

Implementation

Communication